1. Install and configure Ansible on the control node control.example.com as follows:
  - Install the required packages
  - Create a static inventory file called /home/admin/ansible/inventory as follows:
    -- node1.example.com is a member of the dev host group
    -- node2.example.com is a member of the test host group
    -- node3.example.com and node4.example.com are members of the prod host group
    -- node5.example.com is a member of the balancers host group
    -- The prod group is a member of the webservers host group
  - Create a configuration file called ansible.cfg as follows:
    -- The host inventory file /home/admin/ansible/inventory is defined
    -- The location of roles used in playbooks is defined as /home/admin/ansible/roles

2. Create and run an Ansible ad-hoc command:
  As a system administrator, you will need to install software on the managed nodes.
  Create a shell script called yum-pack.sh that runs an Ansible ad-hoc command to create a yum repository on each of t
    1. The name of the repository is EX407
    2. The description is "Ex407 Description"
    3. The base URL is ftp://192.168.10.254/pub/rhel75/
    4. GPG signature checking is enabled
    5. The GPG key URL is ftp://192.168.10.254/pub/rhel75/RPM-GPG-KEY-redhat-release
    6. The repository is enabled

3. Install packages:
  Create a playbook called packages.yml that:
  - Installs the php and mariadb packages on hosts in the dev, test, and prod host groups
  - Installs the Development Tools package group on hosts in the dev host group
  - Updates all packages to the latest version on hosts in the dev host group

4. Use a RHEL system role:
  Install the RHEL system roles package and create a playbook called timesync.yml that:
  -- Runs on all managed hosts
  -- Uses the timesync role
  -- Configures the role to use the time server 192.168.10.254
  -- Configures the role to set the iburst parameter as enabled

5. Create and use a role:
  Create a role called apache in /home/admin/ansible/roles with the following requirements:
  - The httpd package is installed, enabled on boot, and started
  - The firewall is enabled and running with a rule to allow access to the web server
  A template file index.html.j2 exists (you have to create this file) and is used to create the file /var/www/html/index.ht
    Welcome to HOSTNAME on IPADDRESS
  where HOSTNAME is the fully qualified domain name of the managed node and IPADDRESS is the IP address of th
  Create a playbook called httpd.yml that uses this role as follows:
  The playbook runs on hosts in the webservers host group

6. Install roles using Ansible Galaxy:
  Use Ansible Galaxy with a requirements file called /home/admin/ansible/roles/install.yml to download and install rol
  -- http://192.168.10.254/ex407/role1.tar.gz (name: balancer)
  -- http://192.168.10.254/ex407/role2.tar.gz (name: phphello)

7. Create a playbook called balance.yml as follows:

- The playbook contains a play that runs on hosts in the balancers host group and uses the balancer role.
- This role configures a service to load balance web server requests between hosts in the webservers host group.
- When implemented, browsing to hosts in the balancers host group (e.g., http://node5.example.com) should produce:
  Welcome to node3.example.com on 192.168.10.z
  Reloading the browser should return output from the alternate web server:
  Welcome to node4.example.com on 192.168.10.a

8. Create a playbook for phphello role:
   The playbook contains a play that runs on hosts in the webservers host group and uses the phphello role.
   When implemented, browsing to http://<host>/hello.php should produce:
   Hello PHP World from FQDN (where FQDN is the fully qualified domain name of the host)
   along with various details of the PHP configuration including the version of PHP installed.

9. Create a playbook called web.yml as follows:
   - The playbook runs on managed nodes in the dev host group
   - Create the directory /webdev with the following requirements:
     - membership in the apache group
     - regular permissions: owner=read+write+execute, group=read+write+execute, other=read+execute
     - special permissions: set group ID
   - Symbolically link /var/www/html/webdev to /webdev
   - Create the file /webdev/index.html with a single line of text: Development

10. Create and use a partition:
    Create a playbook called partition.yml that runs on all managed nodes and does the following:
    - Creates a single primary partition number 1 of size 1500MiB on device vdb
    - Formats the partition with the ext4 filesystem
    - Mounts the filesystem persistently at /mount1
    - If the requested partition size cannot be created, display "Requested size is not present" and use size 800MiB instea
    - If the device vda does not exist, display "Devices is not present".

11. Create a password vault:
    Create an Ansible vault to store user passwords as follows:
    - Vault name: vault.yml
    - Contains two variables:
      dev_pass with value wakennym
      mgr_pass with value rocky
    - Password to encrypt/decrypt vault: atenorth
    - Password stored in /home/admin/ansible/password.txt

12. Create a playbook called hwreport.yml that produces an output file /root/hwreport.txt on all managed nodes:
    - Information to include:
      -- Inventory host name
      -- Total memory in MB
      -- BIOS version
      -- Size of disk device vda
      -- Size of disk device vdb
    - Each line contains a single key=value pair.
    Your playbook should:
      - Download hwreport.empty from http://192.168.10.254/ex407/hwreport.empty and save it as /root/hwreport.txt
      - Modify with the correct values.
      - If a hardware item does not exist, set the associated value to NONE

13. Generate a hosts file:
- Download initial template hosts.j2 from http://192.168.10.254/ex407/ to /home/admin/ansible/
- Complete the template for /etc/hosts format.

Create a playbook called gen_hosts.yml that uses this template to generate /etc/hosts on hosts in the dev host group.

Resulting /etc/hosts should include:

127.0.0.1 localhost localhost.localdomain localhost4 localhost4.localdomain4
::1 localhost localhost.localdomain localhost6 localhost6.localdomain6
192.168.10.x node1.example.com node1
192.168.10.y node2.example.com node2
192.168.10.z node3.example.com node3
192.168.10.a node4.example.com node4
192.168.10.b node5.example.com node5

14. Modify file content:

Create a playbook called /home/admin/ansible/modify.yml:
- Runs on all inventory hosts
- Replaces the contents of /etc/issue with:

On dev hosts: Development
On test hosts: Test
On prod hosts: Production

15. Rekey an Ansible vault:

Rekey an existing vault:
- Download from http://192.168.10.254/ex407/secret.yml and save as secret.yml
- Current vault password: curabete
- New vault password: newware
- Vault remains encrypted with the new password

16. Create user accounts:

Download user_list.yml from http://192.168.10.254/ex407/user_list.yml to /home/admin/ansible/

Using the existing vault, create a playbook called create_user.yml that:
- Users with job description 'developer': created on dev & test hosts, password from dev_pass variable, supplementary
- Users with job description 'manager': created on prod hosts, password from mgr_pass variable, supplementary group
- Passwords should use the SHA512 hash format.
- Playbook should use the vault password file created elsewhere.